

Name: _____

Date: _____



patterns & sequences



level 3: missing numbers

1 fill in the blanks

Look at each number pattern.
Write the missing numbers in the blanks.

1 4, , 8, , 12

2 20, 18, , 14,

3 , 15, 20, , 30

4 3, 6, , 12,

5 50, , 40, , 30

6 7, , 11, , 15

2 number lines

Look at each number line.
Write the missing numbers.

1 0 10 30

2 5 15 25

3 100 300 500

4 2 4 8

3 missing box

Find the rule in each pattern.
Write the missing number in the box.

1 2 5 8

3 11 21 31

2 9 7 5

4 100 90 80

4 challenge

These patterns are a little trickier.
Write the missing numbers.

1 1, 4, 7, ,

2 24, 21, 18, ,

3 6, 12, 18, ,



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level 2: number patterns

1 continue the pattern

Look at each number pattern.
Write the next 2 numbers in each row.

- | | | | |
|---|-----------------|----------------------|----------------------|
| 1 | 2, 4, 6, 8, | <input type="text"/> | <input type="text"/> |
| 2 | 5, 10, 15, 20, | <input type="text"/> | <input type="text"/> |
| 3 | 1, 3, 5, 7, | <input type="text"/> | <input type="text"/> |
| 4 | 10, 9, 8, 7, | <input type="text"/> | <input type="text"/> |
| 5 | 12, 14, 16, 18, | <input type="text"/> | <input type="text"/> |
| 6 | 30, 25, 20, 15, | <input type="text"/> | <input type="text"/> |

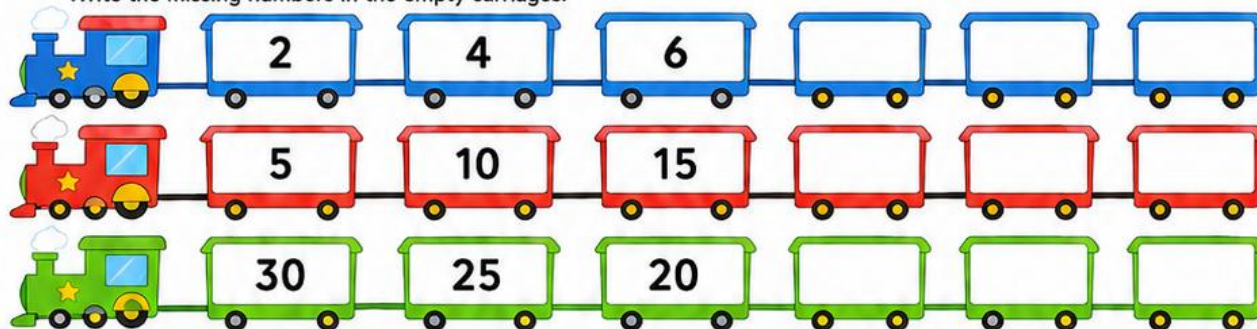
2 find the rule

Look at each number pattern.
Circle the rule that fits.

- | | | | | | |
|---|--------------------|-------|-------|------------|--------|
| 1 | 2, 4, 6, 8, 10 | add 2 | add 5 | subtract 1 | add 10 |
| 2 | 7, 12, 17, 22, 27 | add 2 | add 5 | subtract 1 | add 10 |
| 3 | 20, 19, 18, 17, 16 | add 2 | add 5 | subtract 1 | add 10 |
| 4 | 3, 6, 9, 12, 15 | add 2 | add 3 | subtract 1 | add 10 |

3 fill the number train

Look at the pattern on each train.
Write the missing numbers in the empty carriages.



4 challenge: odd or even?

A Look at each number.
Color ODD numbers red and EVEN numbers blue.

7 ☐ ODD ☐ EVEN	12 ☐ ODD ☐ EVEN	9 ☐ ODD ☐ EVEN	16 ☐ ODD ☐ EVEN	21 ☐ ODD ☐ EVEN	8 ☐ ODD ☐ EVEN	14 ☐ ODD ☐ EVEN	19 ☐ ODD ☐ EVEN
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B Final challenge! Look at the pattern. Write the next 3 numbers.

4, 8, 12, 16, 20,



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level 5: simple nth-term thinking



A sequence follows a rule. The term number tells us which place it is in.

1 look for the rule



Look at each sequence. Fill in the table. Then write what happens each time.

1 2, 4, 6, 8, ...

term number	1	2	3	4
term value	2	4	6	8

What happens each time? _____

2 5, 8, 11, 14, ...

term number	1	2	3	4
term value	5	8	11	14

What happens each time? _____

3 10, 15, 20, 25, ...

term number	1	2	3	4
term value	10	15	20	25

What happens each time? _____

4 1, 3, 5, 7, ...

term number	1	2	3	4
term value	1	3	5	7

What happens each time? _____

5 30, 33, 36, 39, ...

term number	1	2	3	4
term value	30	33	36	39

What happens each time? _____

2 finish the table



Fill in the missing term values.

1

term number	1	2	3	4	5
term value	2	4			

+2
each time

2

term number	1	2	3	4	5
term value	3	6			

+3
each time

3

term number	1	2	3	4	5
term value	5	10	15		

+5
each time

4

term number	1	2	3	4	5
term value	7	9			

+2
each time

3 which rule matches? ✓

Tick the rule that matches each sequence.

1 1, 3, 5, 7, ...

☐ start at 1, add 2 ☐ start at 2, add 2 ☐ start at 1, add 3

2 2, 4, 6, 8, ...

☐ start at 1, add 2 ☐ start at 2, add 2 ☐ start at 2, add 3

3 5, 10, 15, 20, ...

☐ start at 5, add 5 ☐ start at 0, add 5 ☐ start at 5, add 2

4 3, 6, 9, 12, ...

☐ start at 3, add 3 ☐ start at 0, add 3 ☐ start at 3, add 2

4 challenge ★

Find the term.

1 3, 6, 9, 12, ...

What is the 5th term?



2 5, 10, 15, 20, ...

What is the 6th term?

3 2, 4, 6, 8, ...

What is the 5th term?



patterns - sequences

level 4

1 what comes next?

Write the next two numbers in each sequence.

2	,	4	,	6	,	8	,		
30	,	25	,	20	,	15	,		
3	,	6	,	9	,	12	,		
10	,	20	,	30	,	40	,		

2 find the missing number

Work out the rule and fill in the missing number.

4	,	8	,		,	16	,	20
45	,	40	,		,	30	,	25
6	,	12	,	18	,		,	30
100	,		,	80	,	70	,	60

3 growing patterns

Count the objects in each stage. Draw or write how many come next.

a)

1	2	3	4	next
★ ★	★ ★ ★ ★	★ ★ ★ ★ ★ ★	★ ★ ★ ★ ★ ★ ★ ★	

b)

1	2	3	4	next
■	■ ■	■ ■ ■	■ ■ ■ ■	

c)

1	2	3	4	next
●	● ● ●	● ● ● ● ● ●	● ● ● ● ● ● ● ● ● ●	

4 challenge

A pattern starts at 5 and the rule is add 4.

Write the first five numbers:

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level 1: visual patterns

1 copy the pattern

Look at each pattern. Draw the next 2 items.

2 what comes next?

Circle the picture that comes next in each pattern.

1	
2	
3	
4	

3 make your own pattern

Continue each pattern across the row.

4 challenge: is it a pattern?

Look at each row. Is it a repeating pattern? Tick your answer.

1	2	3	4
☐ pattern	☐ pattern	☐ pattern	☐ pattern
☐ not a pattern	☐ not a pattern	☐ not a pattern	☐ not a pattern

